

MH2500 Probability and Introduction to Statistics

Slides for Week 4

Topics: Expectation and Variance

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Review:

- A **random variable** is a function

$$X : \Omega \rightarrow \mathbb{R}.$$

- A **discrete random variable** is a random variable that can only take a finite or at most countably infinite number of values.
- Given a discrete random variable X taking values $x_1, x_2, \dots$, its **probability mass function**, or the **frequency function**, of a random variable X is the function p where

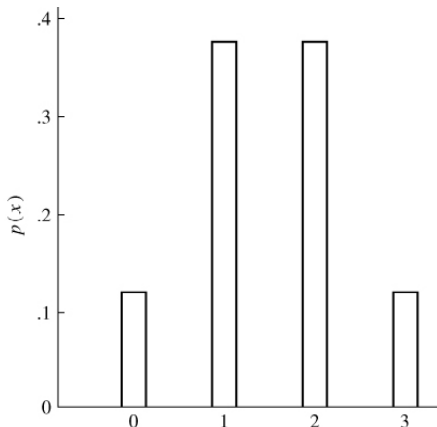
$$p(x_i) = P\{X = x_i\}.$$

- $\sum_{i=1}^{\infty} p(x_i) = 1.$

Example: A coin is flipped three times.

Let X be the total number of heads.

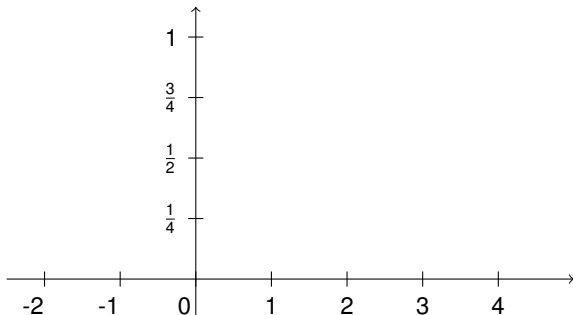
$$p(0) = \frac{1}{8} \quad p(1) = \frac{3}{8} \quad p(2) = \frac{3}{8} \quad p(3) = \frac{1}{8}$$



- The **cumulative distribution function (cdf)** of a random variable is defined as the function $F : \mathbb{R} \rightarrow [0, 1]$ where

$$F(x) = P\{X \leq x\}, \quad -\infty < x < \infty.$$

- We use **uppercase** for cumulative distribution function and **lowercase** letters for probability mass function.

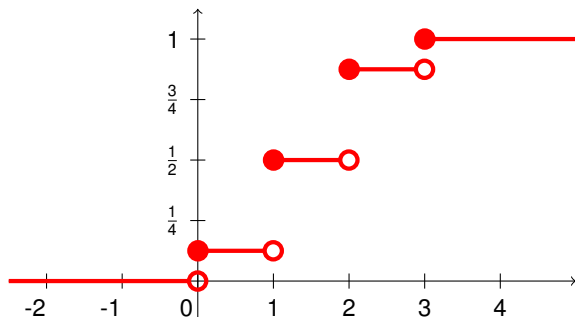


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Note that F is always increasing, right-continuous.

- We also have

$$\lim_{x \rightarrow -\infty} F(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow \infty} F(x) = 1.$$

- Use CDF to find

$$P\{X < a\}, \quad P\{a < X \leq b\}, \quad P\{a \leq X \leq b\}.$$

$$P\{a < X \leq b\} = P\{X \leq b\} - P\{X \leq a\} = F(b) - F(a)$$

$$\begin{aligned} P\{X < a\} &= P\left(\bigcup_{n=1}^{\infty} \left\{X \leq a - \frac{1}{n}\right\}\right) = \lim_{n \rightarrow \infty} P\left\{X \leq a - \frac{1}{n}\right\} \\ &= \lim_{n \rightarrow \infty} F\left(a - \frac{1}{n}\right) = F(a^-). \end{aligned}$$

$$P\{a \leq X \leq b\} = P\{X \leq b\} - P\{X < a\} = F(b) - F(a^-)$$

Expectation

The **expected value of a random variable** is the same as its **weighted average**.

Definition

If X is a discrete random variable with frequency function $p(x)$, the expected value of X , denoted by $\mathbb{E}(X)$, is

$$\mathbb{E}(X) = \sum_i x_i p(x_i),$$

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- $\mathbb{E}(X)$ can also be thought of as **the centre of mass of the frequency function**.

Example 1

A coin is flipped three times. Let X be the total number of heads.

$$\mathbb{E}(X) = 0 \cdot \frac{1}{8} + 1 \cdot \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8} = \frac{3}{2}$$

Example 2

Let X be the outcome of rolling a fair die.

$$\mathbb{E}(X) = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = \frac{7}{2}.$$

Example 3

Say that I is an **indicator** variable for the event A if

$$I = \begin{cases} 1, & \text{if } A \text{ occurs;} \\ 0, & \text{if } A \text{ does not occur} \end{cases}$$

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$$\mathbb{E}(I) = 0 \cdot P\{I = 0\} + 1 \cdot P\{I = 1\} = P\{I = 1\} = P(A).$$

The indicator variable will be very useful, as we will see soon, a couple of weeks later.

Example: Expectation of a Function of a Random Variable

Suppose X is a random variable that takes value in $\{-1, 0, 1\}$ with probabilities

$$P\{X = -1\} = 0.2, \quad P\{X = 0\} = 0.5, \quad P\{X = 1\} = 0.3.$$

Compute $\mathbb{E}(X^2)$.

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Compute $\mathbb{E}(X^2)$.

Let $Y = X^2$. Then

$$P(Y = 0) = 0.5$$

$$P(Y = 1) = 0.2 + 0.3 = 0.5$$

Hence

$$\mathbb{E}(X^2) = \mathbb{E}(Y) = 1 \cdot 0.5 + 0 \cdot 0.5 = 0.5.$$

Note that

$$0.5 = \mathbb{E}(X^2) \neq (\mathbb{E}(X))^2 = 0.01$$

Expectation of a Function of a Random Variable

Let X be a discrete random variable with probability mass function $p(\cdot)$ and $Y = g(X)$. Then

$$\mathbb{E}(Y) = \sum_x g(x)p(x).$$

provided that $\sum_x |g(x)|p(x) < \infty$.

- $Y = g(X)$ is a function of X . As a random variable, X is a function from Ω to $\mathbb{R}$, $g(X)$ is a composition of two functions, and is again a function from Ω to $\mathbb{R}$.
- $p(x) = P\{X = x\}$. As X is a discrete random variable, we can have the range of X as $\{x_1, x_2, \dots, x_n, \dots\}$ and we denote it by A . We thus have $p(x_i) = P\{X = x_i\}$ for each $x_i \in A$. **A is a set of real numbers and can be finite or countably infinite.**

- $\sum_x g(x)p(x)$ is $\sum_{x \in A} g(x)p(x)$, where $p(x) = P\{\omega \in \Omega : P(\omega) = x\}$, and we need to show that

$$\mathbb{E}(Y) = \sum_{x \in A} g(x)p(x).$$

Let B be the range of Y . Then $y \in B$ if and only if $y = g(x_i)$ for some $x_i \in A$. Let $B = \{y_1, \dots, y_m, \dots\}$, and let

$$A_m = \{x \in A : g(x) = y_m\}.$$

- $A_0, A_1, \dots, A_m, \dots$ form a partition of A , the range of X .
- By definition,

$$\mathbb{E}(Y) = \sum_{y \in B} yP\{Y = y\}.$$

Then

$$\begin{aligned} P\{Y = y_j\} &= P\{g(X) = y_j\} = P\{\omega \in \Omega : g(X(\omega)) = y_j\} \\ &= P\{\omega \in \Omega : X(\omega) \in A_j\} = \sum_{x \in A_j} P\{\omega : X(\omega) = x\} = \sum_{x \in A_j} p(x) \end{aligned}$$

and hence

$$\begin{aligned} \mathbb{E}(Y) &= \sum_i y_i P\{Y = y_j\} = \sum_i y_i \sum_{x \in A_i} p(x) = \sum_i \sum_{x \in A_i} y_i p(x) \\ &= \sum_i \sum_{x \in A_i} g(x) p(x) = \sum_{x \in A} g(x) p(x). \end{aligned}$$

This completes the proof.

Corollary

If a and b are constants, then $\mathbb{E}(aX + b) = a\mathbb{E}(X) + b$.

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Proof:

$$\mathbb{E}(aX + b) = \sum_x (ax + b)p(x) = a \sum_x xp(x) + b \sum_x p(x) = a\mathbb{E}(X) + b.$$

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$$\mathbb{E}(X^n) = \sum_x x^n p(x).$$

This is called the **n -th moment** of X . Thus $\mathbb{E}(X)$ is also called the **first moment** of X .

We will see the moment methods later, after the recess week.

Variance

Mean is not powerful enough to characterize a random variable:

$$W = 0 \quad \text{w.p. } 1 \quad Y = \begin{cases} 10, & \text{w.p. } \frac{1}{2} \\ -10, & \text{w.p. } \frac{1}{2} \end{cases} \quad Z = \begin{cases} 100, & \text{w.p. } \frac{1}{2} \\ -100, & \text{w.p. } \frac{1}{2} \end{cases}$$

w.p. — with probability.

New thought: consider **how far X is away from its mean μ** :

- Variance — $\mathbb{E}(|X - \mu|^2)$.

Definition: Variance

If X is a random variable with expected value $\mathbb{E}(X)$, the variance of X is

$$\text{Var}(X) = \mathbb{E}([X - \mathbb{E}(X)]^2)$$

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The standard deviation of X is the square root of the variance.

Let X be the outcome of rolling a fair die. We know $\mathbb{E}(X) = 7/2$. What is $\text{Var}(X)$?

Theorem

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Proof: Let $\mu = \mathbb{E}(X)$. Then

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[(X - \mu)^2] \\&= \sum_x (x^2 - 2x\mu + \mu^2)p(x) \\&= \sum_x x^2 p(x) - 2\mu \sum_x x p(x) + \mu^2 \sum_x p(x) \\&= \mathbb{E}(X^2) - 2\mu \cdot \mu + \mu^2 \\&= \mathbb{E}(X^2) - \mu^2 = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2.\end{aligned}$$

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Hence

$$\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E} X)^2 = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{35}{12}.$$